Formula List for Fiber Optics

1. Numerical Aperture (N.A.)

$$N.A. = \sin\theta_0 = \sqrt{\mu_1^2 - \mu_2^2}$$

Where;

- θ_0 = Acceptance angle
- μ_1 = Refractive index of core
- μ_2 = Refractive index of cladding

2. Numerical Aperture (N.A.) of fiber in other medium

$$N.A. = \sin\theta_0 = \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_o}$$

Where;

- θ_0 = Acceptance angle
- μ_1 = Refractive index of core
- μ_2 = Refractive index of cladding
- μ_o = Refractive index of other outer medium

3. Fractional Refractive Index (Δ)

$$\Delta = \frac{\mu_1 - \mu_2}{\mu_1}$$

Where;

- μ_1 = Refractive index of core
- μ_2 = Refractive index of cladding

4. Numerical Aperture in terms of Δ

$$\mathrm{N.A.} = \mu_1\sqrt{2\Delta}$$

Where;

- μ_1 = Refractive index of core
- Δ = Fractional refractive index of the fibre

5. Critical angle at core-cladding interface (φ_c)

$$\varphi_c = \sin^{-1}\frac{\mu_2}{\mu_1}$$

Where;

- μ_1 = Refractive index of core
- μ_2 = Refractive index of cladding

6. Normalized Frequency (V)

$$V = \frac{2\pi a}{\lambda}\sqrt{{\mu_1}^2 - {\mu_2}^2}$$

Where;

- μ_1 = Refractive index of core
- μ_2 = Refractive index of cladding
- λ = Wavelength travelling through the fiber
- a = Radius of the core

7. Number of modes (N_m)

$$N_m = \frac{V^2}{2}$$

Where;

- V = Normalized Frequency